Do Now · retrieval

Lesson 051 · Linked Lists: Nodes and References

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How does Queue<T> remove items?

Q2

How does a stack differ from a queue?

Q3

What is an object reference?

Q4

A singly linked list is HEAD → node5 → node9 → NULL. A new node7 is available. Describe the two next-pointer assignments needed to insert node7 immediately after node5 without losing access to node9.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Discover — A linked-list node holds a value plus one or more links to neighbouring nodes, rather than occupying an indexed fixed-size array slot.

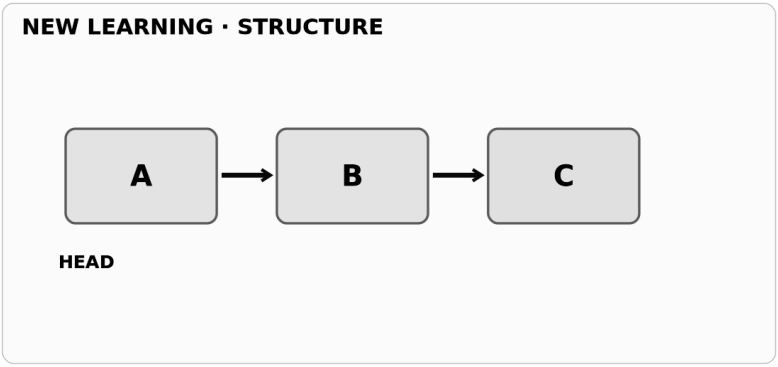
Core idea 2

Trace — C# LinkedList<T> is doubly linked: nodes expose Previous and Next; AddAfter inserts relative to an existing node.

Core idea 3

Evaluate — To insert B between A and C, change logical links to A↔B↔C. The operation differs from shifting array elements.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

node — Value and link information.

Next — Reference to following node.

Previous — Reference to preceding node.

head — First list node.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

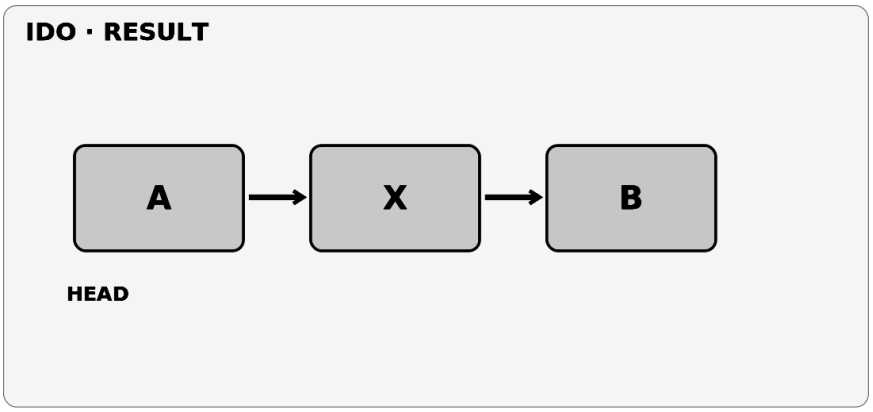
Problem

Trace AddAfter to insert X after A in A→B.

Follow the worked method

1. Explain the trace — New order A,X,B. The inserted node links between A and B.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the trace
P,R remain. After deletion P and R are adjacent logical neighbours.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Draw the logical next-links in HEAD→3→6→NULL after inserting 4 after 3. [6]

Question 2

Write C# using LinkedList<int> to create 2,4 then insert 3 between them. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare middle insertion in linked list with a C# List<int>, including locating position and access. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A linked list always offers O(1) indexed access to its nth node.

Use key vocabulary from memory

1. Explain node in your own words.

2. Explain Next in your own words.

3. Explain Previous in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.